STAFF REVIEW · H446 188 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. No. It scans sequentially and works on unsorted arrays.

2. They must already be sorted.

3. 6.

4. 5: 2 + (8-2)/2 = 5.

I Do — source worked steps

1. Predict first — Use inclusive low and high indexes.

2. Model solution — mid 3 → 15, so low=4; mid 5 → 23, so high=4; mid 4 → 19. Return 4.

We Do — guided CHECK

1. Model solution
mid 4=18, then mid 1=4, then mid 2=9. Return index 2.

You Do — source mark schemes

1. mid=2, match at index 2 on the first comparison.

2. mid 3=8, then mid 5=12, then mid 6=14; the window becomes empty, return -1.

3. With low=mid and only two candidates left, mid may equal low repeatedly; the interval does not shrink, potentially creating an infinite loop.

Exit Ticket — source answer

1. False. It relies on sorted ordering.